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LabTech Testing Limited

BLK A & B, 8/F., HONG KONG SPINNERS INDUSTRIAL BUILDING, PHASE VI, 481-483 CASTLE PEAK ROAD, KOWLOON, HONG KONG
TEL: 2371 0008 FAX: 2744 1037 E-Mail: info@lab-tech.com.hk

WORKSHEET

ULTRASONIC ECHO SOUNDER TEST

Test Method : Hong Kong Concrete Institute TM3 Issue 2 (2024)

Report No.:	**Date of Issue:**	**Page No.: of**
Date of Test:	**Time of Test:**	

General Information (provided by customer)

Customer's Name:	
Customer's Ref. No.:	
Customer's Address:	
Contract No.:	
Works Order No.:	
Project Name:	
Test Location:	**Weather and Temperature: (°C)**

Sample Information (provided by customer)

Pile No.:		**Pile Type:**	
Pile Diameter:	**(mm)**	**Casing Internal Diameter:**	**(mm)**
Existing Ground Level:	**(mPD)**	**Co-ordinate Easting:**	
Platform Ref. Level:	**(mPD)**	**Co-ordinate Northing:**	
Casing Top Level:	**(mPD)**	**Pile Cut-off Level:**	**(mPD)**
Casing Toe Level:	**(mPD)**	**Pile Base Level:**	**(mPD)**
Required Bell-out Diameter:	**(m)**	**Total Pile Length:**	**(m)**
Max. Allowable Inclination:		**Required Rock Socket Dia.:**	**(m)**

Site Conditions of Test Location

Safe Access Provided: Yes ( ) No ( )	**Safe Working Platform Provided: Yes ( ) No ( )**

Setting of Koden Machine

1.5m Resolution (Chart Attached): Yes ( ) No ( )	**Levelling Checked: Yes ( ) No ( )**
Up / Down Speed (m/min):	**Paper Feed Rate:**
Photo of Panel: Yes ( ) No ( )	**Orientation of X: (°)**

Check of Transmission Medium

Type: Water ( ) Bentonite Slurry ( )	**Temperature: / / (°C)**

Test Equipment

1) Koden Machine, I.D. No.:	**5) Caliper, I.D. No.:**
2) Measuring Tape, I.D. No.:	**6) Compass, I.D. No.:**
3) Steel Rule, I.D. No.:	**7) Spirit Level, I.D. No.:**
4) Dipmeter, I.D. No.:	**8) Thermometer, I.D. No.:**

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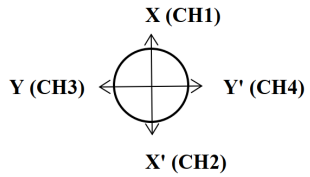
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Verification of Pile Depth

Base level of pile (refer to UES profiles) :	**(mPD)**		
Actual depth measured by weighed tape	**(m)**	**Depth measured from UES profiles**	**(m)**
Difference of Depth (actual - from UES profiles), T	**(m)**		
Criteria: T ≤ 0.3m or ≤ 1% of measured depth	**whichever is smaller**	**( ) Pass**	**( ) Fail**

Pile Orientation and Location

Orientation



Location measured by GPS

Easting: _____

Northing: _____

Tested By: _____

LIN Chung Kit

Checked and Authorize to Release Results By: _____

LAM Hoi Fan / CHU Lap Man

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Test Results

Determination of Horizontal Scaling Factor				
	Channel X-X'		**Channel Y-Y'**	
		Scaling Factor (m/mm)		**Scaling Factor (m/mm)**
First Measurement	**Printed Depth on UES Profiles:**		**Printed Depth on UES Profiles:**	
	(m)		**(m)**	
	Horiz. Reading on UES Profile:		**Horiz. Reading on UES Profile:**	
	(m)		**(m)**	
	Measured Value:		**Measured Value:**	
	(mm)		**(mm)**	
Second Measurement	**Printed Depth on UES Profiles:**		**Printed Depth on UES Profiles:**	
	(m)		**(m)**	
	Horiz. Reading on UES Profile:		**Horiz. Reading on UES Profile:**	
	(m)		**(m)**	
	Measured Value:		**Measured Value:**	
	(mm)		**(mm)**	
Third Measurement	**Printed Depth on UES Profiles:**		**Printed Depth on UES Profiles:**	
	(m)		**(m)**	
	Horiz. Reading on UES Profile:		**Horiz. Reading on UES Profile:**	
	(m)		**(m)**	
	Measured Value:		**Measured Value:**	
	(mm)		**(mm)**	
	Average Scaling Factor (*S*) (m/mm)		**Average Scaling Factor (*S*) (m/mm)**	

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Test Results (Cont'd)

Pile No.		
Rock Socket Diameter	**Channel X-X'**	**Channel Y-Y'**
At the Level (mPD)		
Printed Depth on UES Profile (m)		
Measurement on UES Profile (mm)		
Average Scaling Factor (*S*) (m/mm)		
Calculated Rock Socket Diameter (m)		
Bell-out Diameter	**Channel X-X'**	**Channel Y-Y'**
At the Level (mPD)		
Printed Depth on UES Profile (m)		
Measurement on UES Profile (mm)		
Average Scaling Factor (*S*) (m/mm)		
Calculated Bell-out Diameter (m)		
Verticality Check (Overall)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		

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Test Results (Cont'd)

Pile No.		
Verticality Check (1st Section)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		

Verticality Check (2nd Section)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		

Verticality Check (3rd Section)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		

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Test Results (Cont'd)

Pile No.		
Verticality Check (4th Section)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		

Verticality Check (5th Section)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		

Verticality Check (6th Section)	**Channel X-X'**	**Channel Y-Y'**
Measurement from (mPD)		
to (mPD)		
Depth between Two Measurements (m)		
Offset from the Reference Line (*L*) (mm)	- =	- =
Average Scaling Factor (*S*) (m/mm)		
Calculated Displacement (=*L* **S*) (m)		
Inclination (1 : *V*)		
Towards		
Resultant Displacement (m)	[( ) ^{2} + ( ) ^{2}] $^{1/2}$ =	
Resultant Inclination (1 : *V*)		